

END OF TERM ONE EXAMINATION

S. 5 APPLIED MATHEMATICS

Time: *2 hours*

Attempt all items

Item 1

A school administers a Mathematics test marked out of **100%** to a group of **80 students** in preparation for a regional competition. The teacher records the scores (in percentages) and wants to analyse the overall performance before deciding whether to organise extra coaching sessions.

The recorded scores are:

45, 52, 60, 67, 72, 55, 49, 80, 76, 68,
54, 62, 71, 59, 64, 73, 77, 82, 48, 50,
66, 69, 74, 78, 81, 83, 58, 61, 65, 70,
75, 79, 84, 85, 53, 57, 63, 67, 72, 76,
80, 82, 86, 88, 90, 47, 51, 56, 60, 62,
68, 71, 74, 77, 79, 81, 83, 85, 87, 89,
91, 92, 93, 94, 46, 48, 52, 55, 59, 63,
66, 70, 73, 75, 78, 80, 82, 84, 86, 88

The school policy states:

- If the **median score is below 60%**, extra coaching must be organised.
- If the **middle 50% of students score above 65%**, the class is considered to be performing well.
- The school further states that if the mean score is lower than the median score, the class performance is considered to be affected by low-scoring students.

Task: You have been called as a mathematics student to help the school in;

- (a) Organizing the scores into a frequency table using class intervals of width 10 starting from 40–49,
- (a) deciding whether extra coaching is required
- (b) determining whether the class performance meets the school's expectation class.
- (c) Use the mean and median scores to advise the teacher on the overall performance of the class.

Item 2

During a safety assessment exercise near Lake Victoria, a team of trainee divers is being evaluated before being allowed to participate in a deep-water recovery operation.

A safety officer must decide whether the group is ready for deployment. According to safety guidelines, divers are considered adequately prepared if the typical underwater endurance time is at least 12 minutes.

The recorded underwater times (in minutes) for the trainees are:

15, 8, 7, 20, 10, 11, 24, 12, 13, 35, 12 11

One senior diver argues that the group is ready because the average time exceeds 12 minutes, while another insists that most divers are not ready and that the average may be misleading.

1. Determine the median time.
2. Determine the mode of the data.
3. Calculate the mean time.

END